

Project Proposal: The Relationship Between Shipping Activity, Air Pollution, and Public Health in Southern California Port Communities

Southern California contains some of the highest shipping activity in the United States. Ports such as the Port of Los Angeles provide critical infrastructure for trade and shipping activity. While necessary, these ports do rely heavily on fossil fuel energy and contribute heavily to air pollution due to their greenhouse gas emissions from cargo ships, freight trucks, and port operations. Our project is motivated by a desire to understand the relationship between environmental hazards caused by Southern California port activity and the nearby communities. These environmental hazards represent a public health issue, especially to those living near these ports. We are interested in examining how these environmental hazards may affect these surrounding communities and how public data can help us analyze these relationships and potential disparities. Therefore, our research question is whether higher shipping activity near major Southern California ports is associated with higher levels of air pollution and worse health outcomes in nearby communities.

We plan to use two main public datasets for this project. The first dataset is the Port of Los Angeles shipping statistics dataset, which provides information about shipping activity, cargo volume, and container traffic in Southern California ports. In particular, we plan to use container traffic statistics and annual port traffic data to measure shipping activity near major ports.

<https://www.portoflosangeles.org/business/statistics>

The second dataset is CalEnviroScreen, a California environmental health dataset that contains information about pollution exposure and environmental hazards in communities across California. It includes measurements such as PM2.5 levels, diesel pollution, traffic pollution, asthma prevalence, and other measures of community exposure.

<https://oehha.ca.gov/calenviroscreen>

These datasets are suitable for our project because they directly support our research question. We want to investigate whether higher shipping activity near Southern California ports is associated with higher levels of air pollution in nearby communities. The Port of Los Angeles dataset allows us to measure shipping activity, while CalEnviroScreen provides pollution exposure indicators that can be used to measure environmental impacts on surrounding communities. These two datasets are publicly available and can be downloaded from official government and environmental health websites. Overall, these

datasets allow us to analyze the possible relationship between port activity and air pollution in Southern California communities.

For our chosen datasets, we need to take into account that while the shipping data is continuous, the CalEnviroScreen is not, and offers periodic survey data spread out over years. Both are extremely large datasets: the Port of LA data provides a concise time series that gives us access to seasonal changes and exact volume, and CalEnviroScreen gives us health and importantly location data for those surveyed. Thus we can model the health effect given distance to the Port of LA as well as activity at given times. This would call for multivariate linear regression, where we model a specific health problem within CalEnviroScreen (say Asthma diagnosis), and how it can be predicted by exposures to different pollutants, shipping traffic, and proximity to the Port of LA. Another plus of our chosen data is that CalEnviroScreen's geographical data allows us to potentially create spatial analysis by showing the severity of health hazards associated with close proximity to the port. This would require our knowledge of ANOVA to test the statistical significance of our expected higher health problems close to the shipping traffic.

Our expected outcomes would most likely be that higher shipping activity near major Southern California ports does contribute to higher air pollution levels in nearby communities. As stated in previous parts, the shipping activities around Southern California ports do rely on energy that would cause air pollution a lot, such as fossil fuel. Burning fossil fuel would release nitrogen oxide, which is a gas that contributes to the formation of acid rain and smog, thus significantly reducing the level of air quality. According to data, the main sources of nitrogen oxide emissions include ships. Even though part of the nitrogen oxide emitted would be absorbed by water, it turns out that water is only able to absorb a small portion of the gas, leaving the rest of gas in the air. In addition, the absorbed nitrogen oxide combined with water produces nitric acid and nitrous acid. Nitric acid evaporates at room temperature, meanwhile nitrous acid is unstable and will decompose back into nitrogen and water, both resulting in acid rain. Therefore, we infer that air pollution increases in communities near ports with heavy shipping activities.